# AI110 | Foundations of AI Engineering

Foundations of AI Engineering Summer 2026 (@ Section 1A | Tuesday 5PM - 7PM PT)
Personal Member ID#: **151098**



Overview | Tinker (Lab) | Show (Project) | Report

📬 Reminder: Project 3 is due by **Monday, July 20th at 1:59AM CDT.** Submit

# Show What You Know: Music Recommender Simulation

# ℹ️ Project Overview

⏰ **~4 hours**

You are working for a startup music platform that wants to understand how big-name apps like Spotify or TikTok predict what users will love next. Your mission is to simulate and explain how a basic music recommendation system works by designing a modular architecture in Python that transforms song data and "taste profiles" into personalized suggestions.

## 🎯 Goals

By completing this project, you will be able to...

- [ ] Explain the transformation of data into predictions, distinguishing between input features, user preferences, and ranking algorithms.
- [ ] Design and implement a weighted-score recommender in Python that uses attributes like genre, mood, and energy to calculate song relevance.
- [ ] Identify and document algorithmic bias, using AI to brainstorm how simple content-based systems might create "filter bubbles" or over-favor certain genres.
- [ ] Communicate technical reasoning through a structured Model Card that details intended use, data limitations, and future improvements.

# 🪛 Project Instructions

## ▾ Phase 1: Understanding the Problem

⏰ **~25 mins**

Before you build your own recommender, you'll first explore how real platforms like Spotify or TikTok decide what to suggest next. In this phase, you'll identify what data these systems rely on, how they represent "taste," and sketch how a simple recommendation process might work.

**Step 1: Explore Real Recommendation Systems**

- [ ] Go to the [Music Recommender Simulation] repo.
- [ ] Click **Fork** to create your own copy under your GitHub account.
- [ ] Clone the fork to your computer, then open the cloned folder in VS Code.
- [ ] Use your AI coding assistant's chat to research and summarize how major streaming platforms (like Spotify or YouTube) predict what users will love next. Structure your prompt to specifically ask for the difference between collaborative filtering (using other users' behavior) and content-based filtering (using song attributes).

- [ ] Identify the main data types involved in these systems, such as likes, skips, playlists, tempo, or mood.

**Step 2: Identify Key Features**

- [ ] Examine the `data/songs.csv` file to see the available attributes for your simulator, such as `genre`, `mood`, `energy`, and `tempo_bpm`.
- [ ] Attach `songs.csv` and ask your AI coding assistant to analyze the available data and suggest which features would be most effective for a simple content-based recommender. Evaluate if the suggested features (e.g., energy, valence) align with your personal experience of how a musical "vibe" is defined.

**Step 3: Mapping the Logic**

- [ ] Determine your "Algorithm Recipe"—the set of rules your system will use to score songs.
- [ ] Formulate a prompt to help you design a math-based "Scoring Rule" for your recommender. Ask your AI coding assistant how to calculate a score for a numerical feature (like energy) that rewards songs that are closer to the user's preference, rather than just having higher or lower values.

  > Think about the weights. Should a matching genre be worth more points than a matching mood?

- [ ] Ask your AI coding assistant to explain why we need both a "Scoring Rule" (for one song) and a "Ranking Rule" (for a list of songs) to build a recommendation system.

**Step 4: Summarize Your Concept**

- [ ] Open `README.md`.
- [ ] In the **How The System Works** section, write a short paragraph explaining your understanding of how real-world recommendations work and what your version will prioritize.
- [ ] List the specific features your `Song` and `UserProfile` objects will use in your simulation.

📍 **Checkpoint**: You have a clear concept sketch of how a recommender connects user data to song data, plus a written explanation of how real systems might do this at scale.

Close Section

## ▾ Phase 2: Designing the Simulation

⏰ **~45 mins**

Now that you understand how music recommendations work conceptually, it's time to design your own simplified version. In this phase, you'll plan what data your recommender will use and sketch out how it will calculate which songs to suggest before writing any real code.

**Step 1: Define Your Data**

- [ ] Open the `data/songs.csv` file in your project. This is your initial catalog of 10 songs.
- [ ] Review the features for each song, such as `genre`, `mood`, and `energy` (on a 0.0–1.0 scale).
- [ ] Use your AI coding assistant's chat to help expand this dataset. Formulate a prompt that asks the AI to generate 5–10 additional songs in a valid CSV format that includes the existing headers. Ensure the new songs represent a diverse range of genres and moods not already present in the starter file.

You can also ask your AI coding assistant to suggest new numerical features, like "Danceability" or "Acousticness," to add more depth to your simulation.

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**Step 2: Create a User Profile**

- [ ] Define a specific "taste profile" that your recommender will use for its comparisons.
- [ ] This profile should be a dictionary containing target values for the features you identified in Step 1 (e.g., `favorite_genre`, `favorite_mood`, `target_energy`).
- [ ] Use your AI coding assistant's chat to get a critique of your proposed user profile. Structure your prompt to ask if these specific preferences will allow the system to differentiate between "intense rock" and "chill lofi," or if the profile is too narrow.

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**Step 3: Sketch the Recommendation Logic**

- [ ] Describe your "algorithm recipe"—the specific rules your program will use to decide which songs to recommend.
- [ ] Open a **new chat session** for "Scoring Logic Design." Attach `songs.csv` and ask your AI coding assistant for point-weighting strategies. Your prompt should seek a balance: for example, how much should a "Mood" match count compared to a "Genre" match?
- [ ] Finalize your recipe. A common starting point is:
  - **+2.0 points** for a genre match.
  - **+1.0 point** for a mood match.
  - **Similarity points** based on how close the song's energy is to the user's target.

**Step 4: Visualize the Design**

- [ ] Sketch a quick map of the data flow to plan your logic: **Input (User Prefs) → Process (The Loop: Judging every individual song in the CSV using your scoring logic) → Output (The Ranking: Top K Recommendations).**

> This is just a planning aid for *you* — it isn't graded, so use whatever format is fastest: a napkin sketch, a Mermaid flowchart, a whiteboard photo, or a few lines of text. The goal is to make sure you understand how a single song moves from the CSV to a ranked list before you start coding.

**Step 5: Document Your Plan**

- [ ] Open `README.md`.
- [ ] Instead of using separate files, document your plan in the **How The System Works** section.
- [ ] Include your finalized "Algorithm Recipe" and a brief note on any potential **biases** you expect (e.g., "This system might over-prioritize genre, ignoring great songs that match the user's mood").

📍 **Checkpoint**: You have a clearly defined plan for your recommender, including an expanded dataset, a specific user profile, and a weighted scoring logic. You are now ready to begin the implementation phase.

Close Section

## ▾ Phase 3: Implementation

⏰ **~90 mins**

In this phase, you'll turn your design into real Python code. You'll build the functions that load your song data, score each track, and generate a ranked list of recommendations. Along the way, you'll use AI tools to brainstorm, debug, and compare versions while making sure *you* stay in control of the logic.

**Step 1: Set Up Your Project Files**

- [ ] Open `src/recommender.py`. This is where your core logic will live.

- [ ] Use **agent mode** or your AI coding assistant's chat to implement the `load_songs` function. Your prompt should instruct the AI to use Python's `csv` module to read `data/songs.csv` and return a list of dictionaries.

  > Make sure your prompt specifies that numerical values (like `energy` or `tempo_bpm`) must be converted to floats or integers so you can do math with them later.

- [ ] Verify your progress by running the `main()` function in `src/main.py`. It should print out "Loaded songs: 10" (or however many you have in your CSV).

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**Step 2: Implement the Scoring Function**

- [ ] In `src/recommender.py`, find the `score_song(user_prefs, song)` function.
- [ ] Highlight the function and use your AI coding assistant's chat. Formulate a prompt based on your "Algorithm Recipe" from Phase 2. Tell the AI how many points to award for a genre match, a mood match, and how to calculate a score for numerical features like energy.

  > To earn full credit on the rubric, ensure your scoring logic returns both a numeric score and a list of "reasons" (e.g., "genre match (+2.0)") so the user understands the recommendation

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**Step 3: Build the Recommender Function**

- [ ] In `src/recommender.py`, find the `recommend_songs(user_prefs, songs, k)` function.

  > Recommending is simply the act of ranking. To find the "best" songs, this function must use your `score_song` function as a "judge" for every single in the catalog. Once every song has a numeric score, you can then sort the entire list to find the top results.

- [ ] Attach `recommender.py` and use your AI coding assistant's chat. Ask for the most "Pythonic" way to loop through all songs, calculate their scores using your new function, and return the top k results sorted from highest to lowest score.

  > Ask the AI to explain the difference between using `.sort()` and `sorted()` so you understand how your data is being handled.

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**Step 4: CLI Verification**

- [ ] Open `src/main.py`.

- [ ] Use your AI coding assistant's chat to format the output of your recommendations. Ask for a clean, readable layout in the terminal that displays the song title, the final score, and the specific "reasons" generated by your scoring function.
- [ ] Run the script: `python -m src.main`. Verify that the top results match what you would expect for the default "pop/happy" profile.
- [ ] In your `README.md`, add a section called **Sample Recommendation Output** and paste the terminal output showing the recommendations (song titles, scores, and reasons) into a **fenced code block** (not a screenshot).

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**Step 5: Document and Commit**

- [ ] Use your AI coding assistant to add 1-line docstrings to your new functions in `recommender.py`.
- [ ] Ask your AI coding assistant to generate a commit message to summarize your implementation. Make sure your message mentions that you have a working "CLI-first" simulation.
- [ ] Push your changes: `git push origin main`.

📍 **Checkpoint**: You now have a working Python recommender in `src/recommender.py` that loads data from a CSV, computes scores based on user preferences, and produces a ranked, explained list of suggestions in your terminal.

Close Section

## ▾ Phase 4: Evaluate and Explain

⏰ **~45 mins**

With your recommender up and running, it's time to investigate how well it works and why it behaves the way it does. In this phase, you'll test your system with different user profiles, examine the strengths and weaknesses of your scoring logic, and practice explaining the results clearly.

**Step 1: Stress Test with Diverse Profiles**

- [ ] Open `src/main.py`.
- [ ] Define at least three distinct user preference dictionaries (e.g., "High-Energy Pop," "Chill Lofi," "Deep Intense Rock").

- [ ] Open a new chat session for "System Evaluation." Reference or attach the relevant files and ask your AI coding assistant to suggest "adversarial" or "edge case" user profiles—profiles designed to see if your scoring logic can be "tricked" or if it produces unexpected results (e.g., a user with conflicting preferences like energy: 0.9 and mood: sad).
- [ ] Run your recommender for each profile and observe the top 5 results in your terminal.
- [ ] Paste the terminal output for each profile's recommendations as **fenced code blocks** in your `README.md` or `model_card.md` (not screenshots).

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**Step 2: Look for Accuracy and Surprises**

- [ ] Compare the recommendations for at least one profile to your own musical intuition. Do the results "feel" right?
- [ ] Use your AI coding assistant's chat on a specific result in your terminal (or the corresponding code in `main.py`). Formulate a prompt asking your AI coding assistant to explain why a specific song ranked first based on your current weights in `recommender.py`.

  > If the same song keeps appearing at the top of every list, your "Genre" weight might be too strong, or your dataset might be too small to provide variety.

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**Step 3: Run a Small Data Experiment**

- [ ] Choose one change to test your system's sensitivity:
  - **Weight Shift:** Double the importance of energy and half the importance of genre.
  - **Feature Removal:** Temporarily comment out the mood check to see how the rankings change.
- [ ] Use your AI coding assistant to quickly apply one of these experimental changes across `recommender.py`. In your prompt, describe the specific logic change you want to see and ask it to verify that the math remains valid.
- [ ] Run your `main.py` again. Note whether the change made the recommendations more accurate or just different.

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**Step 4: Identify Bias and Limitations**

- [ ] Attach `recommender.py` and `songs.csv` and ask your AI coding assistant to identify potential "filter bubbles" or biases in your current scoring logic. For example, does your system ignore certain types of users because of how you calculate the "energy gap"?
- [ ] Open `model_card.md`.

- [ ] In the **Limitations and Bias** section, write 3–5 sentences describing one weakness you discovered during your experiments (e.g., "The system over-prioritizes pop because 60% of the dataset is pop music").

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**Step 5: Document Your Evaluation**

- [ ] Update the **Evaluation** section of your `model_card.md`.
- [ ] Describe which user profiles you tested and what surprised you about the results.
- [ ] For each pair of profiles, write at least one comment in your `model_card.md` file comparing the differences between their outputs — what changed, and why does it make sense? For example: *"EDM profile prefers high energy songs; acoustic profile shifts toward low energy guitars."* This helps demonstrate that you understand what your user preferences are actually testing for and whether the output is valid.

  > Use plain language. Imagine you are explaining to a non-programmer why the "Gym Hero" song keeps showing up for people who just want "Happy Pop."

📍 **Checkpoint**: You have tested your recommender with multiple diverse profiles, run at least one logic experiment, and identified a clear limitation or bias. You have documented these findings in your `model_card.md`.

Close Section

# ▾ Phase 5: Reflection and Model Card

> ⏰ **~25 mins**
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> In this final phase, you'll step back and reflect on what your recommender does well, where it struggles, and what this tells you about real-world AI systems. You'll turn your findings into a simple Model Card, a common industry practice for documenting how an AI system works, its intended use, and its limitations

**Step 1: Fill Out the Model Card Template**

- [ ] Complete each section of the `model_card.md` file in your project repo. Use short, simple sentences! This is meant to be clear, not formal.

> **Model Card Sections**
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> 1. **Model Name:** Choose something fun but descriptive (e.g., "VibeFinder 1.0").
> 2. **Goal / Task:** Explain what your recommender tries to predict or suggest.

3. **Data Used:** Describe your dataset size, features, and any limits.
4. **Algorithm Summary:** Explain your scoring rules in plain language (not code).
5. **Observed Behavior / Biases:** Describe at least one pattern, limitation, or imbalance.
6. **Evaluation Process:** Summarize how you tested your system (profiles, experiments, comparisons).
7. **Intended Use and Non-Intended Use:** Clarify what your system is *designed* for and what it shouldn't be used for.
8. **Ideas for Improvement:** List 2-3 things you'd change if you kept developing this.

**Step 2: Write a Personal Reflection**

- [ ] In the final section of your Model Card (or the `README.md`), write a personal reflection on your engineering process.
  - [ ] What was your biggest learning moment during this project?
  - [ ] How did using AI tools help you, and when did you need to double-check them?
  - [ ] What surprised you about how simple algorithms can still "feel" like recommendations?
  - [ ] What would you try next if you extended this project?

📍 **Checkpoint**: You've created a complete Model Card documenting how your recommender works, identified key limitations and biases, and written a thoughtful reflection on what you learned and how AI shaped your process.

Close Section

## ▾ Optional Extensions

⏰ **~30 mins**

These add-ons are completely optional if you want to deepen or personalize your recommender. Pick one or try them all!

- [ ] **Challenge 1: Add Advanced Song Features**
  - Introduce **5 or more** complex attributes to your dataset that are not currently present in the baseline data, such as **Song Popularity (0-100)**, **Release Decade**, or **Detailed Mood Tags** (e.g., "nostalgic," "aggressive," "euphoric").
  - Update both `data/songs.csv` and the scoring logic in `src/recommender.py` so scoring accounts for the new attributes.
  - In `ai_interactions.md`, document the agentic workflow: the example prompt(s) you used, a summary of the changes the AI generated, and your manual verification notes.
- [ ] **Challenge 2: Create Multiple Scoring Modes**
  - Build two or more different ranking strategies (e.g., "Genre-First," "Mood-First," or "Energy-Focused") so a user can switch between modes in `main.py`.
  - Attach `recommender.py` and use your AI coding assistant's chat to brainstorm a design pattern (like a simple "Strategy" pattern) that keeps your code modular.
  - In your `ai_interactions.md`, document the chosen design pattern and how AI contributed to brainstorming it.
- [ ] **Challenge 3: Diversity and Fairness Logic**
  - Implement a "Diversity Penalty" that prevents the recommender from suggesting too many songs from the same artist or genre in the top results.
  - Formulate a prompt for your AI coding assistant's chat that describes a rule to penalize a song's score if its artist is already present in the top recommendations list.
- [ ] **Challenge 4: Visual Summary Table**
  - Improve the readability of your terminal output by providing a formatted table or summary.
  - Ask your AI coding assistant to suggest a way to use a library like `tabulate` or simple ASCII formatting to display your top recommendations. Ensure your prompt specifies that the table must include the "reasons" for each score.

📍 **Checkpoint**: You've gone beyond the core requirements and added meaningful complexity, new features, or deeper analysis to your recommender.

Close Section

## 📬 Submitting Your Project

Once you've completed all the required features for your project, use the following checklist to prepare your work for submission.

1. Code is pushed to the correct GitHub repository

2. Repo is public
3. Required files are present (code, README, reflection, tests if applicable)
4. Commit history shows multiple meaningful commits
5. Reflection answers the prompts with specific, honest details
6. Final changes are committed and pushed before the deadline
7. Submit your assignment using the submit button.

## 🗺️ How It's Graded

*A detailed breakdown of graded features and points can be found on the course grading page.*